

Problem Statement and Research Plan

Project: LLM Evaluation Analytics Dashboard

1. What problem am I solving?

I am addressing the problem of efficiently evaluating and comparing the performance of multiple Large Language Models (LLMs). The goal is to develop an interactive dashboard that can collect, evaluate, compare, and visualize LLM responses using multiple quality and performance metrics, helping developers, researchers, and organizations understand which models perform better for different tasks.

2. Why is this problem important?

This problem is important because LLMs can produce different results for the same prompt, and manually evaluating a large number of responses is time-consuming, costly, and difficult to manage consistently. Simple scores alone may also fail to explain why one model performs differently from another.

Solving this problem can improve evaluation efficiency, transparency, model selection, and decision-making for developers, researchers, and organizations building AI applications.

3. What has already been done?

Several approaches have already been used for LLM evaluation, including static benchmark datasets, human evaluation, automated evaluation metrics, dynamic evaluation, interactive evaluation, and LLM-as-a-Judge methods.

Existing evaluation frameworks generally assess aspects such as correctness, relevance, factuality, coherence, safety, and response quality. Research has also explored visual and interactive approaches for comparing LLM responses and analysing evaluation results.

4. What are the limitations of existing approaches?

Despite their usefulness, existing approaches may have limitations such as:

- Evaluation results being distributed across different tools and methods
- Heavy dependence on manual human evaluation
- Difficulty in comparing multiple LLMs in one place
- Limited interactive exploration of large evaluation datasets
- Difficulty understanding why models perform differently
- Possible bias or inconsistency in LLM-as-a-Judge evaluation
- Limited combined analysis of quality, cost, latency, and safety

- Difficulty in continuously tracking model performance

Therefore, there is a need for an integrated system that can combine automated evaluation, multi-model comparison, and interactive visual analytics in one platform.

5. What exactly will my system do?

The proposed system will evaluate and compare multiple LLMs using standardized prompts or datasets and present the results through an interactive analytics dashboard.

It will take user-defined prompts or evaluation datasets as input and send them to selected LLMs through APIs. Their responses will be collected and evaluated using methods such as LLM-as-a-Judge and predefined evaluation metrics. The results will then be processed and visualized through charts, graphs, tables, filters, and comparison panels.

Main Objectives

- Evaluate multiple LLMs using common prompts and datasets.
- Compare model performance across quality, safety, efficiency, and cost-related metrics.
- Visualize evaluation results through an interactive dashboard.
- Identify strengths, weaknesses, errors, and performance patterns across different models.

The system will be designed to make LLM evaluation more centralized, scalable, interpretable, and easier to use.

6. How will I evaluate whether my solution is better or useful?

The proposed system will be evaluated using performance, reliability, efficiency, and usability metrics.

The evaluation may include:

- **Evaluation Agreement** – Compare automated/LLM-based evaluation with human judgments where ground-truth ratings are available.
- **Evaluation Consistency** – Check whether repeated evaluations produce stable results.
- **Response Quality Metrics** – Measure factors such as accuracy, relevance, factuality, coherence, and safety.
- **Latency and Processing Time** – Measure how quickly evaluations are completed.
- **Cost and Token Usage** – Track the resources required for model generation and evaluation.
- **Comparison with Existing Methods** – Compare dashboard-based evaluation with basic manual or separate evaluation approaches.
- **Usability Evaluation** – Assess how easily users can understand, filter, compare, and interpret the dashboard results.

The solution will be considered successful if it provides reliable and interpretable evaluation results, reduces manual evaluation effort, supports effective model comparison, and helps users make informed decisions about LLM selection and improvement.